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In [1]: import numpy as np
import pandas as pd

# 1. Create a DataFrame from a List
data = [
    [1, 'Smith', 50000],
    [2, 'Jones', 60000]
]

df = pd.DataFrame(data)
df.columns = ['Empd', 'Name', 'Salary']

print("Employee DataFrame:")
print(df)

# 2. Display DataFrame information
print("\nDataFrame Information:")
df.info()

# 3. Create an employee dataset
employee_data = {
    'emp id': [1, 2, 3, 4, 5, 6, 7],
    'name': [
        'SREE VARSSINI K S',
        'SREEMATHI B',
        'SREYA G',
        'SREYASKARI MULLAPUDI',
        'SRI AKASH U G',
        'SRI HARSHAVARDHANAN R',
        'SRI HARSHAVARDHANAN R'
    ],
    'salary': [5000, 6000, 7000, 5000, 8000, 3000, 6000]
}

df = pd.DataFrame(employee_data)

# 4. Display the first and last five rows
print("\nFirst Five Rows:")
print(df.head())

print("\nLast Five Rows:")
print(df.tail())

# 5. Display DataFrame information
print("\nEmployee DataFrame Information:")
df.info()

# 6. Select the salary column
print("\nSalary Column:")
print(df['salary'])

# 7. Calculate statistical measures
print("\nMean Salary:", df['salary'].mean())
print("Median Salary:", df['salary'].median())
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print("Mode Salary:", df['salary'].mode().tolist())
print("Salary Variance:", df['salary'].var())
print("Salary Standard Deviation:", df['salary'].std())

# 8. Display numerical summary
print("\nStatistical Summary:")
print(df.describe())

# 9. Display summary of all columns
print("\nComplete Summary:")
print(df.describe(include='all'))

# 10. Display column names
print("\nColumn Names:")
print(df.columns)

# 11. Convert DataFrame into a NumPy array
emp_array = df.values
print("\nEmployee Array:")
print(emp_array)

# 12. Create a DataFrame from the NumPy array
employee_DF = pd.DataFrame(emp_array, columns=df.columns)

print("\nFinal Employee DataFrame:")
print(employee_DF)
```

Employee DataFrame:

	Empd	Name	Salary
0	1	Smith	50000
1	2	Jones	60000

DataFrame Information:

<class 'pandas.DataFrame'>

RangeIndex: 2 entries, 0 to 1

Data columns (total 3 columns):

#	Column	Non-Null Count	Dtype
0	Empd	2 non-null	int64
1	Name	2 non-null	str
2	Salary	2 non-null	int64

dtypes: int64(2), str(1)

memory usage: 190.0 bytes

First Five Rows:

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000

Last Five Rows:

	emp id	name	salary
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

Employee DataFrame Information:

<class 'pandas.DataFrame'>

RangeIndex: 7 entries, 0 to 6

Data columns (total 3 columns):

#	Column	Non-Null Count	Dtype
0	emp id	7 non-null	int64
1	name	7 non-null	str
2	salary	7 non-null	int64

dtypes: int64(2), str(1)

memory usage: 410.0 bytes

Salary Column:

0	5000
1	6000
2	7000
3	5000
4	8000
5	3000
6	6000

Name: salary, dtype: int64

Mean Salary: 5714.285714285715

Median Salary: 6000.0
Mode Salary: [5000, 6000]
Salary Variance: 2571428.5714285714
Salary Standard Deviation: 1603.5674514745463

Statistical Summary:

	emp id	salary
count	7.000000	7.000000
mean	4.000000	5714.285714
std	2.160247	1603.567451
min	1.000000	3000.000000
25%	2.500000	5000.000000
50%	4.000000	6000.000000
75%	5.500000	6500.000000
max	7.000000	8000.000000

Complete Summary:

	emp id	name	salary
count	7.000000	7	7.000000
unique	NaN	6	NaN
top	NaN	SRI HARSHAVARDHANAN R	NaN
freq	NaN	2	NaN
mean	4.000000	NaN	5714.285714
std	2.160247	NaN	1603.567451
min	1.000000	NaN	3000.000000
25%	2.500000	NaN	5000.000000
50%	4.000000	NaN	6000.000000
75%	5.500000	NaN	6500.000000
max	7.000000	NaN	8000.000000

Column Names:

Index(['emp id', 'name', 'salary'], dtype='str')

Employee Array:

```
[[1 'SREE VARSSINI K S' 5000]
 [2 'SREEMATHI B' 6000]
 [3 'SREYA G' 7000]
 [4 'SREYASKARI MULLAPUDI' 5000]
 [5 'SRI AKASH U G' 8000]
 [6 'SRI HARSHAVARDHANAN R' 3000]
 [7 'SRI HARSHAVARDHANAN R' 6000]]
```

Final Employee DataFrame:

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

In []: